

Bellabeat Smart Device Usage Analysis

SLEEP • STEPS • CALORIES • BMI • HEART RATE

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Objective

- ▶ Analyze smart device usage to understand user activity patterns.
- ▶ Identify trends in steps, calories, and engagement.
- ▶ Provide actionable recommendations to improve user experience.

Dataset Overview

- ▶ Data Source: Bellabeat smart device usage dataset
- ▶ Period: March to May
- ▶ Key metrics tracked: BMI, Weight, Steps, Calories Burned, Sleep Patterns

Methodology

- ▶ Data Cleaning – Removing duplicates, handling missing values
- ▶ Data Transformation – Creating measures and KPIs in DAX
- ▶ Visualization – Power BI dashboards for trends and insights
- ▶ Analysis – Understanding activity levels, peak usage, and correlations

Average Active Tracking Days

- ▶ Average Active Tracking Days per User: 39 days.
- ▶ This suggests that, on average, users engage with their Bellabeat device for around 5–6 weeks.
- ▶ It could indicate short- to medium-term engagement, meaning users may not consistently track their activity throughout the entire period.

39

Avg Active Tracking Days
per user

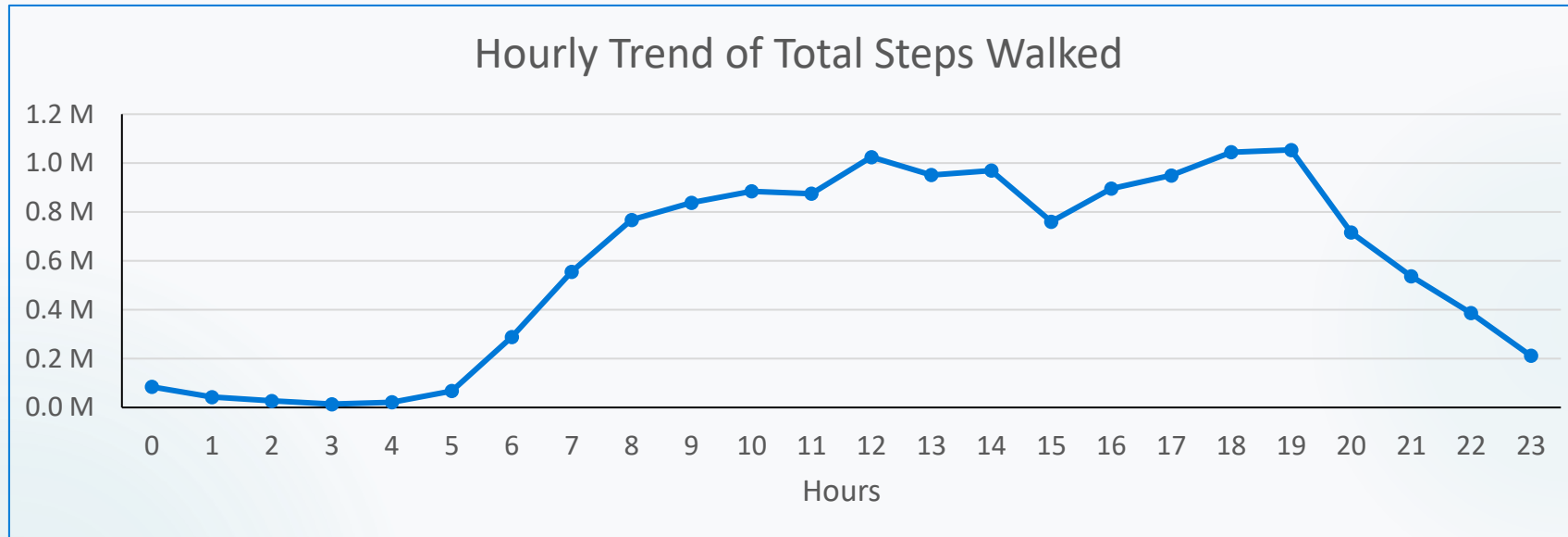
User Activity Patterns

- ▶ Users collectively recorded 10M steps in the analysis period.
- ▶ Average steps per day: ~164K total across all users.
- ▶ This indicates high engagement with Bellabeat devices, showing users are actively tracking their daily movement.

10M
Total Steps

164K
Avg Total Steps

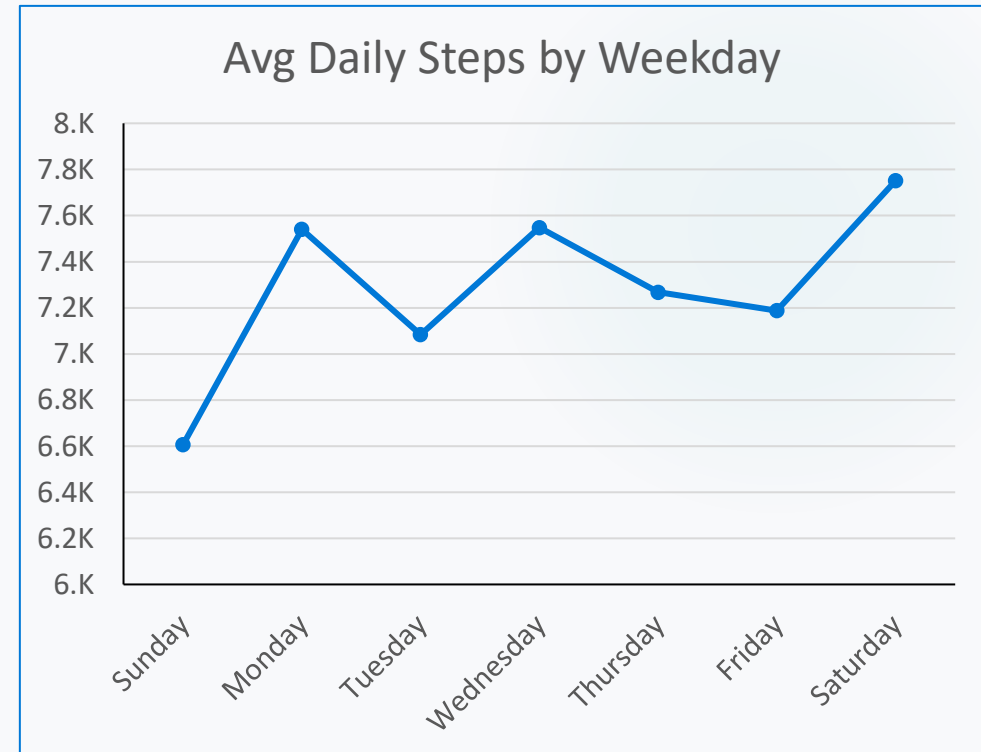
Hourly Walking Activity Trends



- ▶ Step activity peaks twice → 12 PM–1 PM and 6 PM–7 PM.
- ▶ Lowest activity between 12 AM– 4 AM.
- ▶ Steps rise from 5 AM and drop after 7 PM.

Weekly Trends in Average Steps Walked

- ▶ Users are most active on Wednesdays and Saturdays with the highest average daily steps (~7.8K).
- ▶ Monday also shows a strong start to the week with higher-than-average steps.
- ▶ Lowest activity occurs on Sunday (~7K steps), indicating a potential rest day for many users.



Average Daily Calorie Consumption

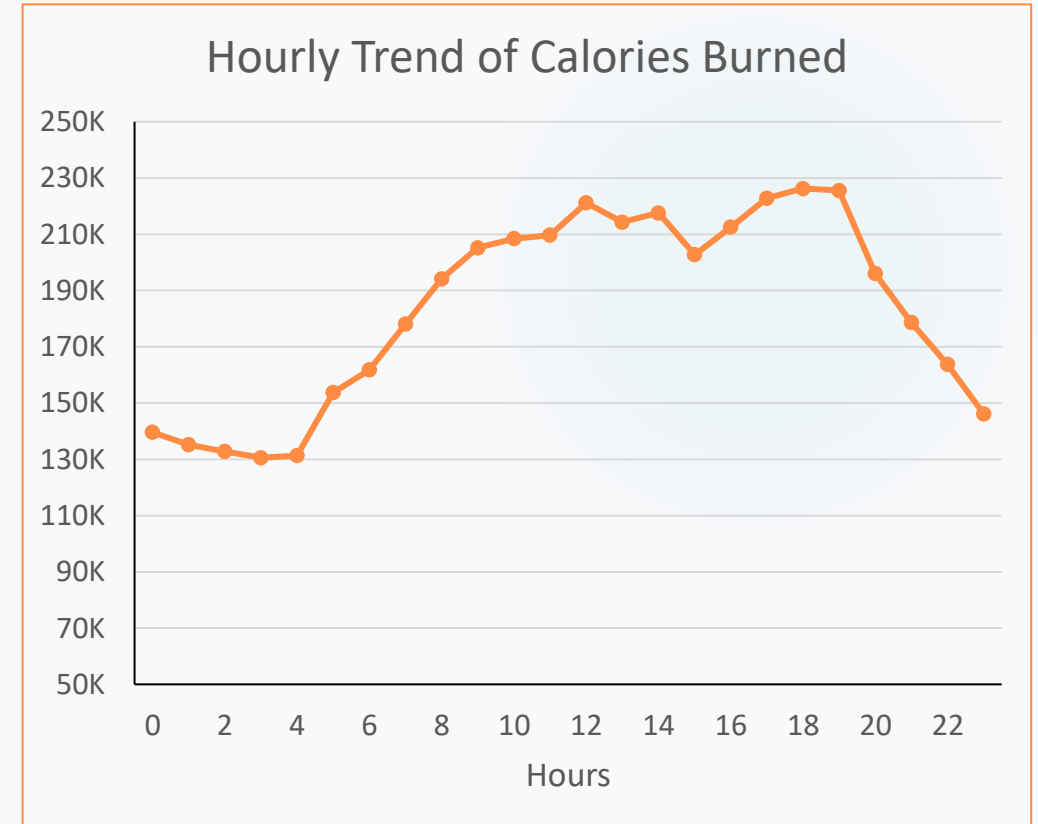
- ▶ Average Calories Burned per Day: 51,000 calories (51K) across all users.
- ▶ If we break this down per user, the average daily burn is approximately 1,600–2,000 calories, which aligns with typical healthy activity levels.
- ▶ This value is consistent with the step count and activity patterns we've analyzed earlier.

51K

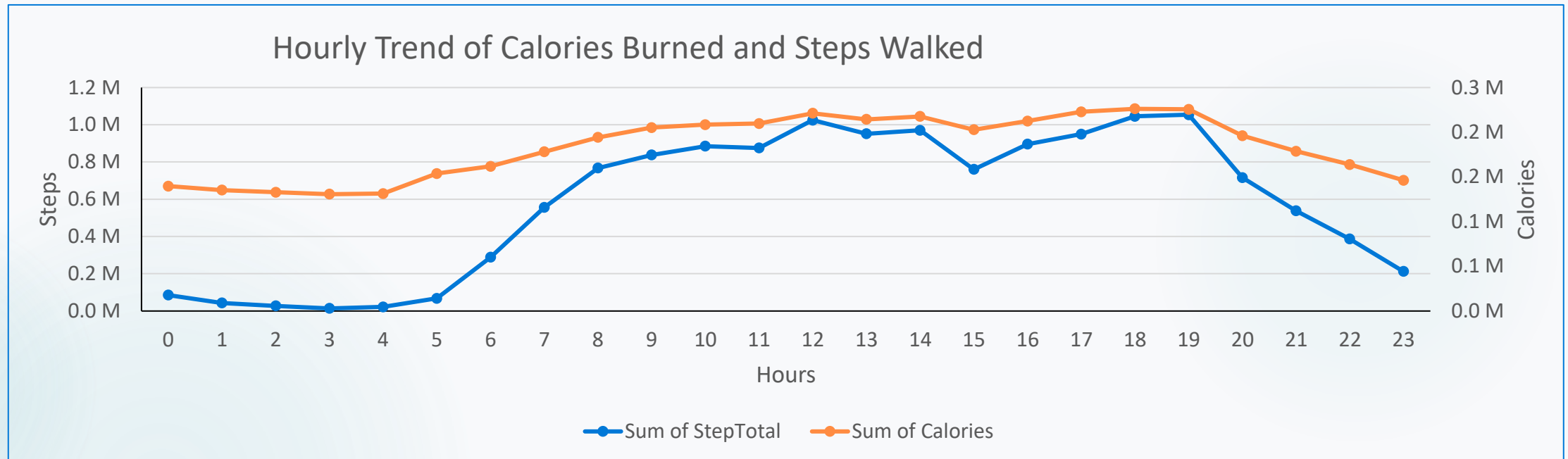
Avg Total
Calories/Day

Hourly Calorie Burned Trend

- ▶ Peak Calorie Burn Hours:
 - 12 AM (221K calories)
 - 6 PM to 7 PM (226K calories) → highest calorie burn
 - 12 PM & 1 PM (~218K each) also remain high.
- ▶ Calorie burn ramps up after 7 AM, peaks during late morning & evening workout hours, and drops sharply after 7 PM.



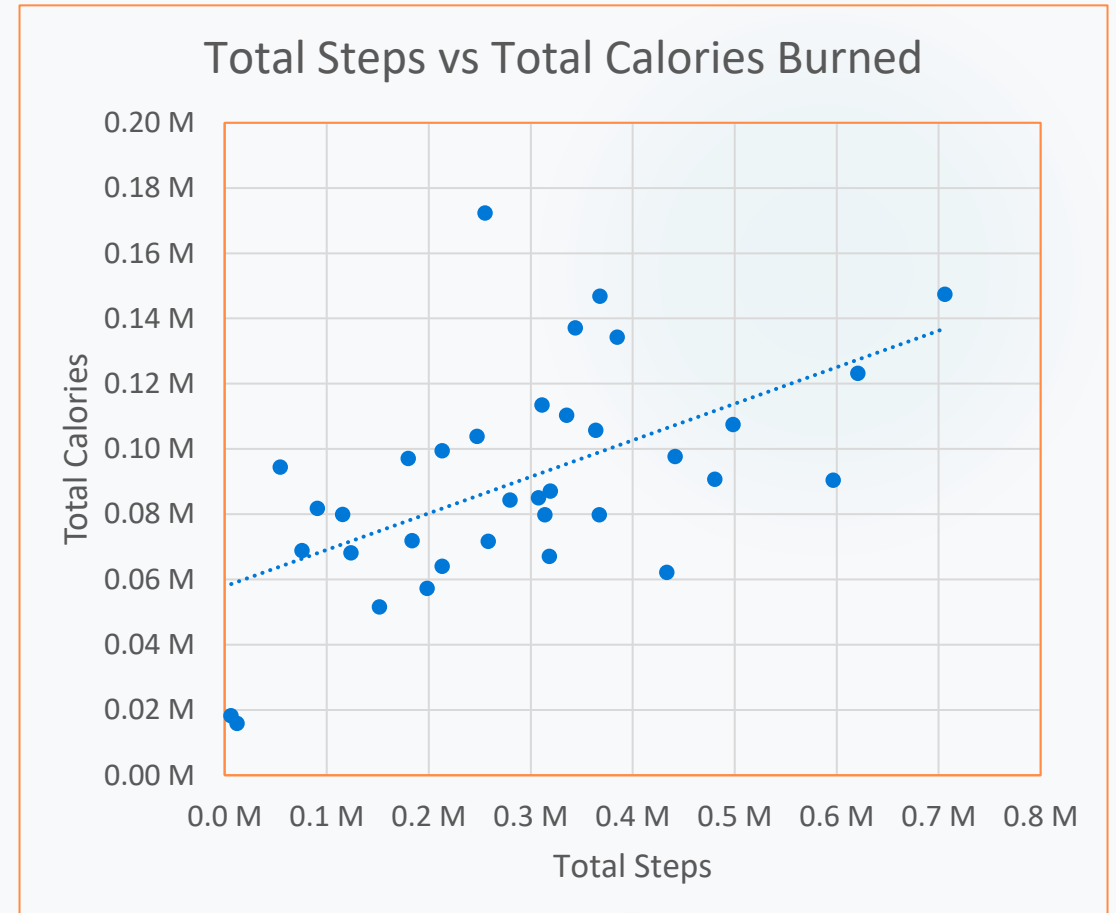
Steps vs Calories Correlation



- ▶ As steps increase, calories burned also increase almost proportionally.
- ▶ Users who walk more consistently burn significantly more calories.
- ▶ Bellabeat devices are effectively capturing daily activity intensity.

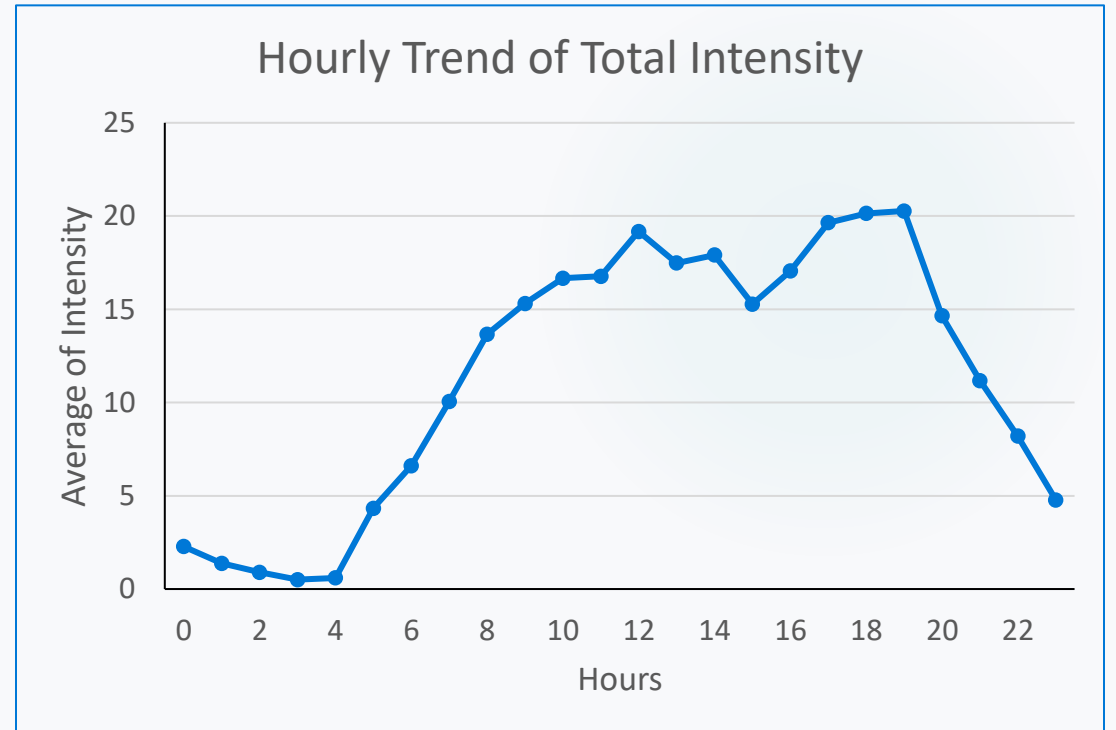
Steps vs Calories Correlation

- ▶ Both the line chart and scatter plot confirm that more steps = more calories burned.
- ▶ Most users cluster between 100K – 400K steps and 60K – 120K calories.
- ▶ Higher step counts (>500K) are associated with significantly higher calorie burn.
- ▶ There are no extreme outliers, meaning calorie burn scales consistently with steps.
- ▶ Encouraging users to increase step activity can directly improve calorie expenditure and overall health.



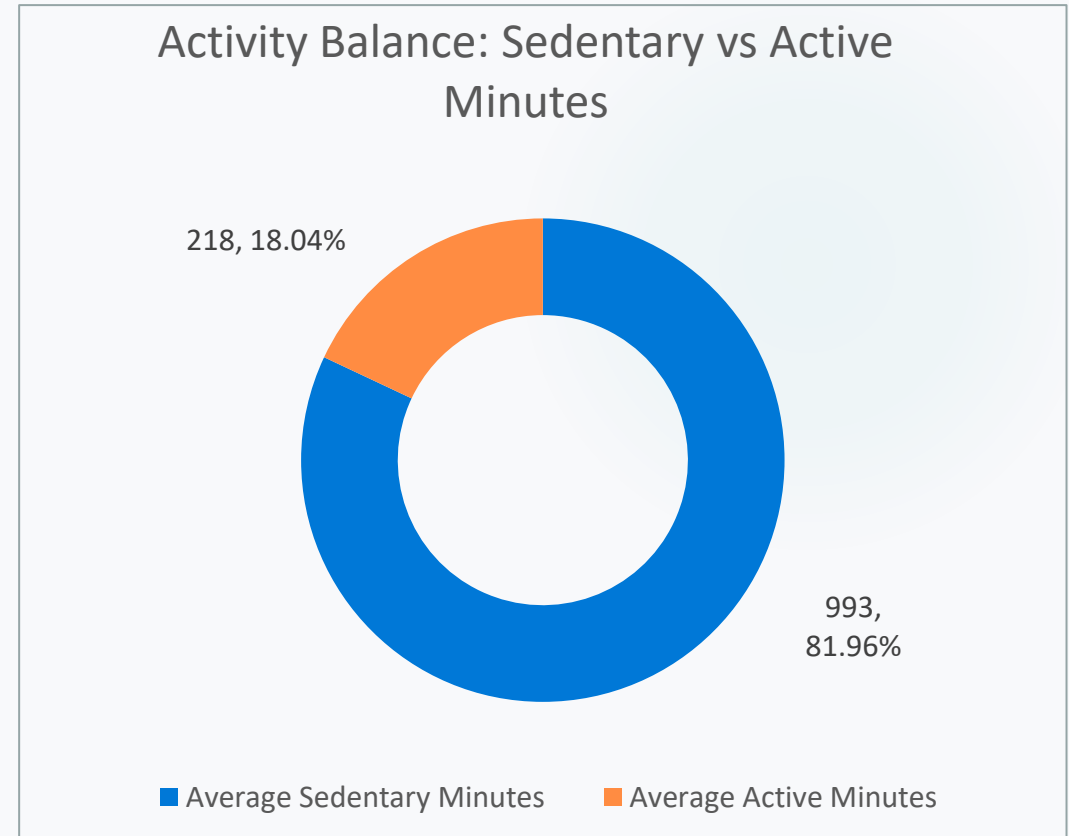
Hourly Activity Intensity Trends

- ▶ Two activity peaks: Late morning (~12 AM) and Evening (~7 PM).
- ▶ Lowest activity happens between midnight and early morning.
- ▶ There's a post-lunch dip (2 PM – 4 PM), suggesting possible sedentary periods.



Active vs Sedentary Time Analysis

- ▶ Average Sedentary Minutes: 993 min/day. 81.96% of the day is spent inactive.
- ▶ Average Active Minutes: 218 min/day. 18.04% of the day involves physical activity.
- ▶ This shows a significant imbalance between sedentary and active time.
- ▶ Users spend over 4X more time being sedentary than active.
- ▶ Despite occasional activity peaks, most users lead a sedentary lifestyle.



Average Daily Sleep Duration

- ▶ Average Sleep Duration: 6.66 hours per night.
- ▶ This is below the recommended 7–9 hours of sleep for adults.
- ▶ Indicates that users may not be getting optimal rest, which could affect their overall wellness and activity recovery.

6.66

Avg Sleep
Duration

Sleep Duration Overview

- ▶ Total Time in Bed: 386K minutes
- ▶ Total Time Asleep: 355K minutes
- ▶ This indicates that, on average, users spend more time in bed than actually sleeping.
- ▶ Sleep Efficiency = 92%
- ▶ 92% sleep efficiency → good overall sleep quality, but there's still a gap of ~31K minutes spent awake in bed.

386K

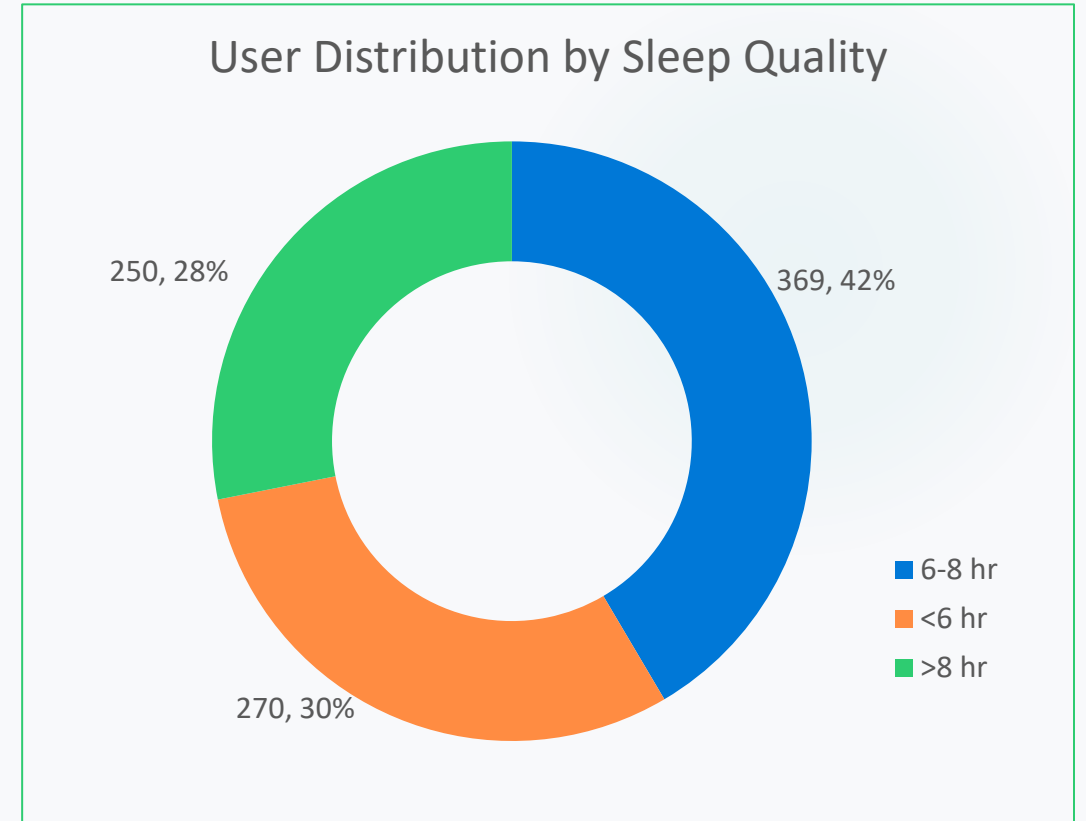
Total Time in Bed

355K

Total Minutes
Asleep

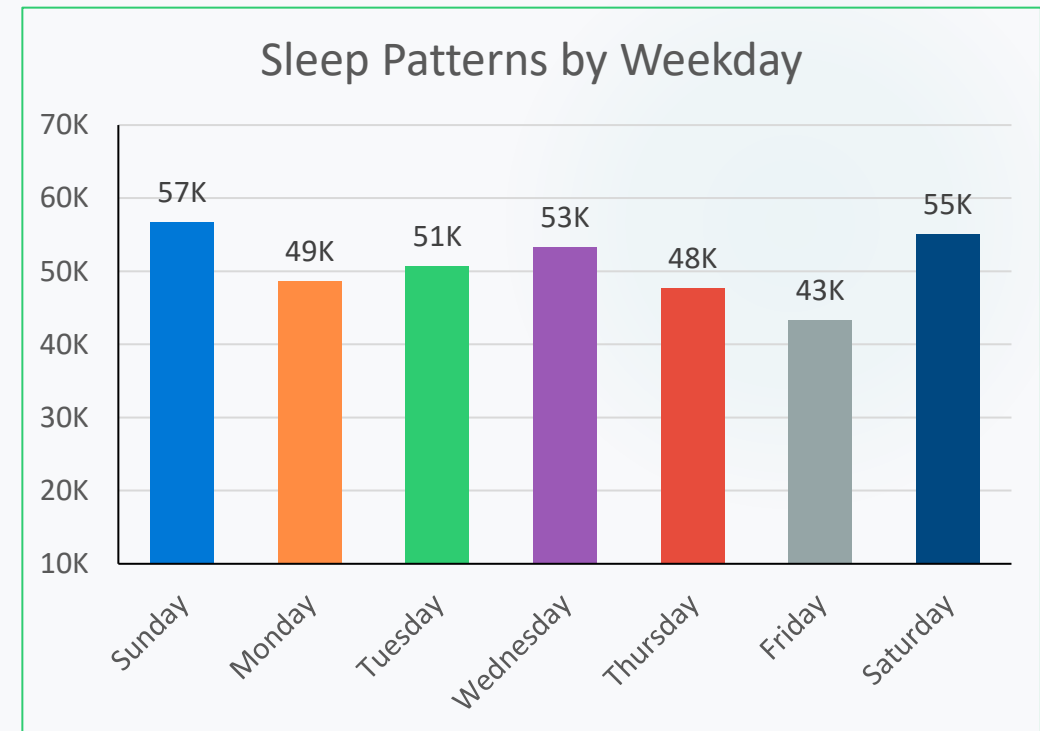
Sleep Quality Distribution

- ▶ 42% of users (369 users) sleep 6–8 hours → Healthy sleep range
- ▶ 30% of users (270 users) sleep < 6 hours → Potential sleep deprivation
- ▶ 28% of users (250 users) sleep > 8 hours → Indicates longer rest periods but could also signal irregular sleep schedules.



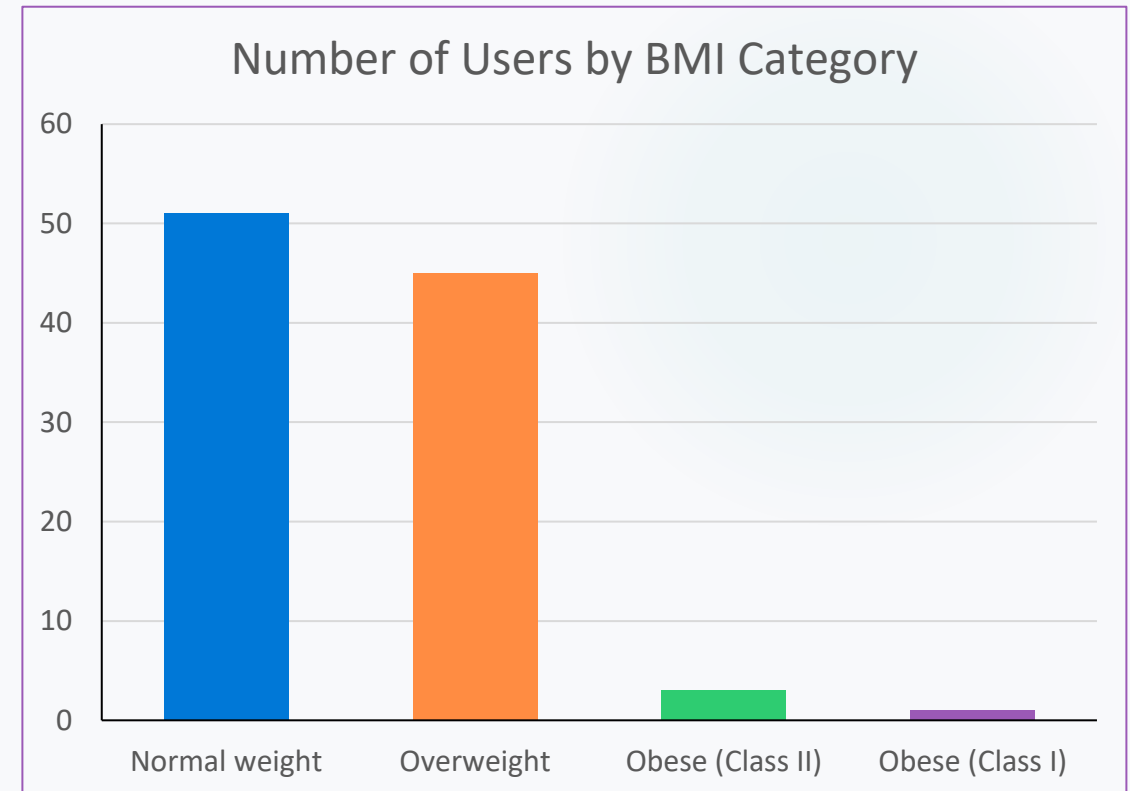
Sleep Patterns Across Weekdays

- ▶ Highest sleep duration: Users tend to sleep more on weekends.
- ▶ Moderate sleep duration: Tuesday (51K), Wednesday (53K), and Monday (49K) indicate average rest periods.
- ▶ Lowest sleep duration: Friday (43K mins) → Users sleep the least, possibly due to late-night social activities or weekend routines.



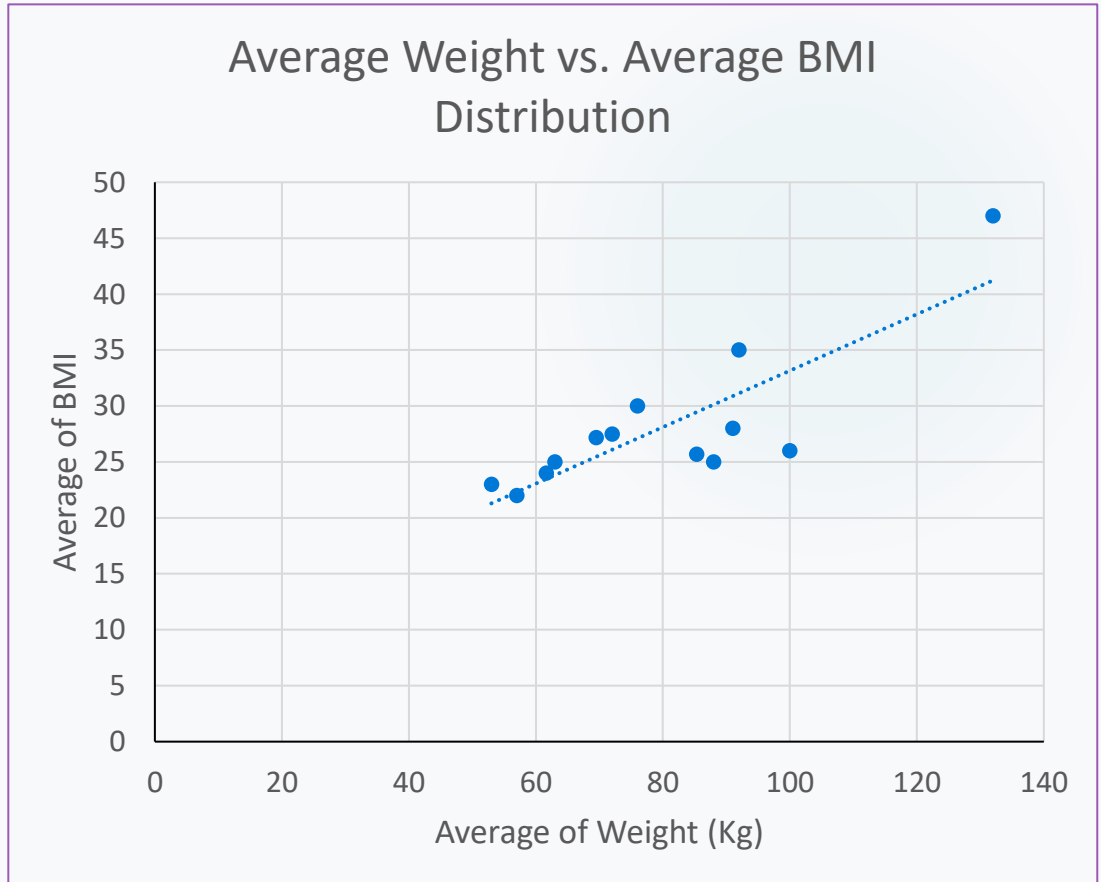
BMI Category Distribution

- ▶ Normal Weight: 51% of users fall into the healthy BMI range, showing good weight management.
- ▶ Overweight: 45% of users are above the ideal BMI, indicating potential lifestyle improvement opportunities.
- ▶ Obese (Class I & II): Less than 4% of users fall into obesity categories, which is relatively low but still relevant for targeted guidance.



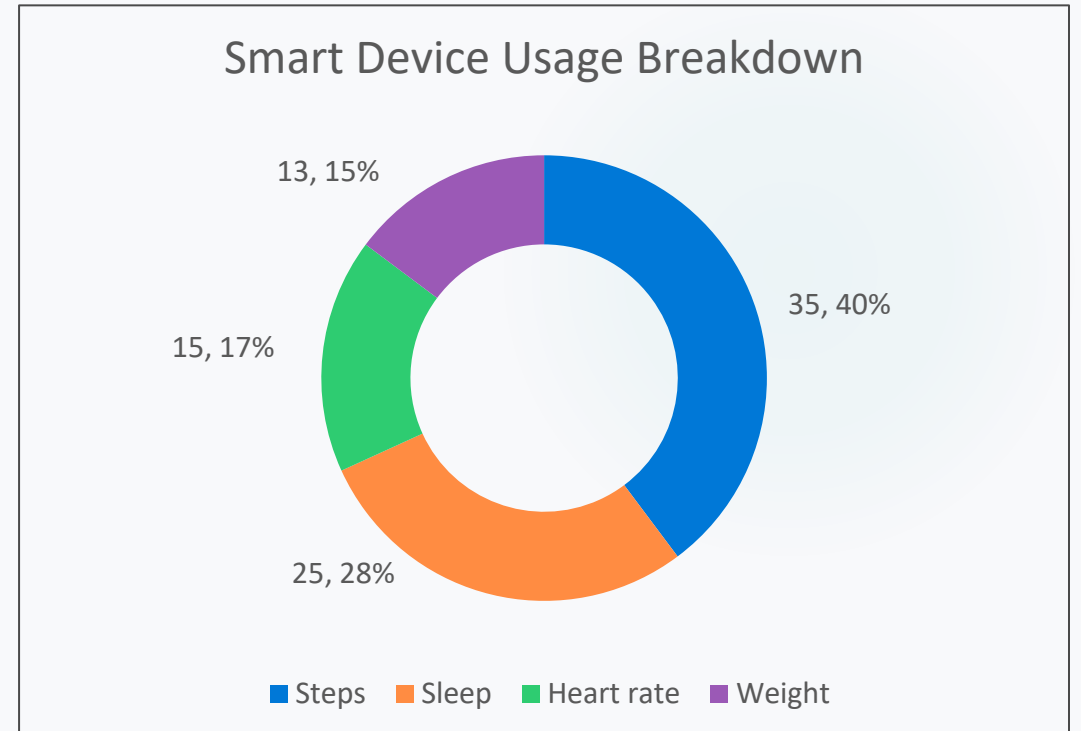
Weight vs BMI Correlation

- ▶ There's a positive correlation between weight and BMI — as weight increases, BMI generally rises.
- ▶ Most users have a weight range between 50–90 kg and a BMI range between 20–30, which is close to the healthy range.
- ▶ A few outliers with high weight (>100 kg) also have BMI > 35, indicating possible obesity.
- ▶ The dotted trend line clearly shows a consistent upward relationship.



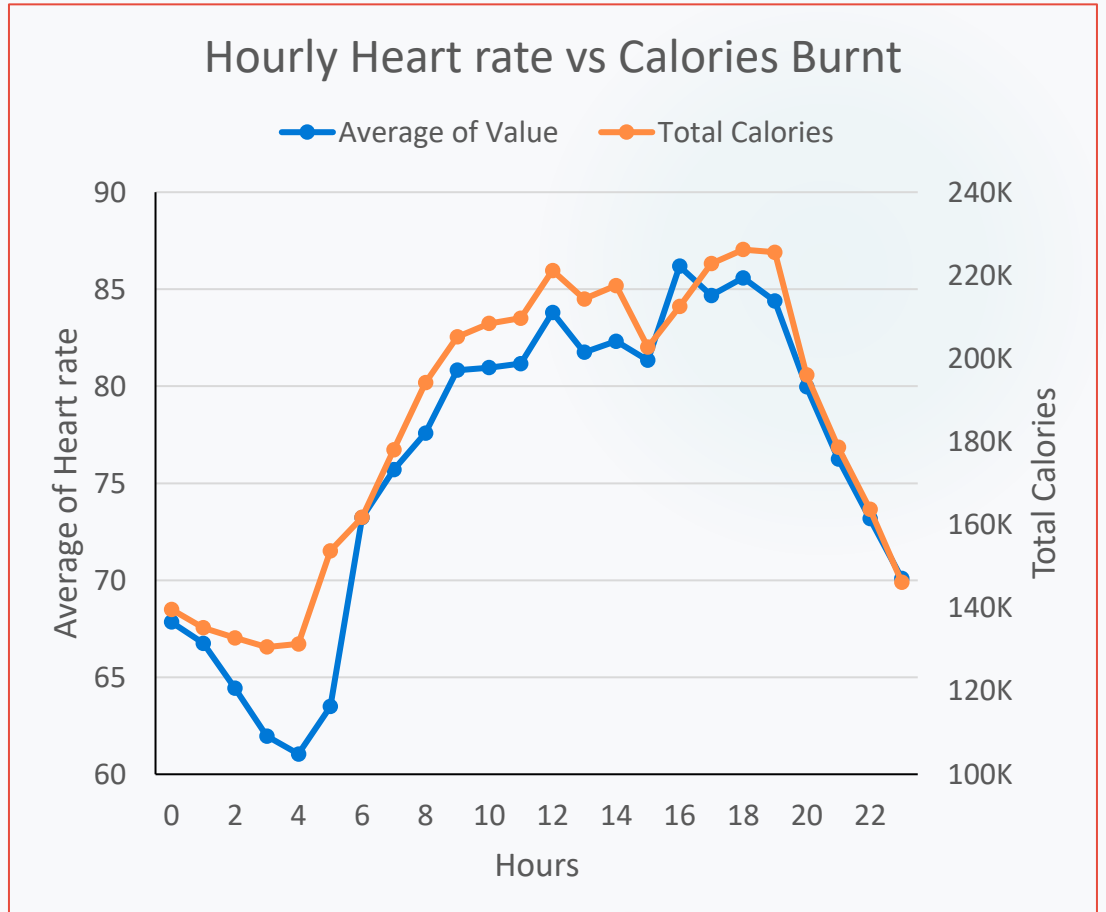
Purpose of Device Usage

- ▶ 40% use it mainly for step tracking.
- ▶ 28% for sleep tracking.
- ▶ 17% for heart rate monitoring.
- ▶ 15% for weight tracking.



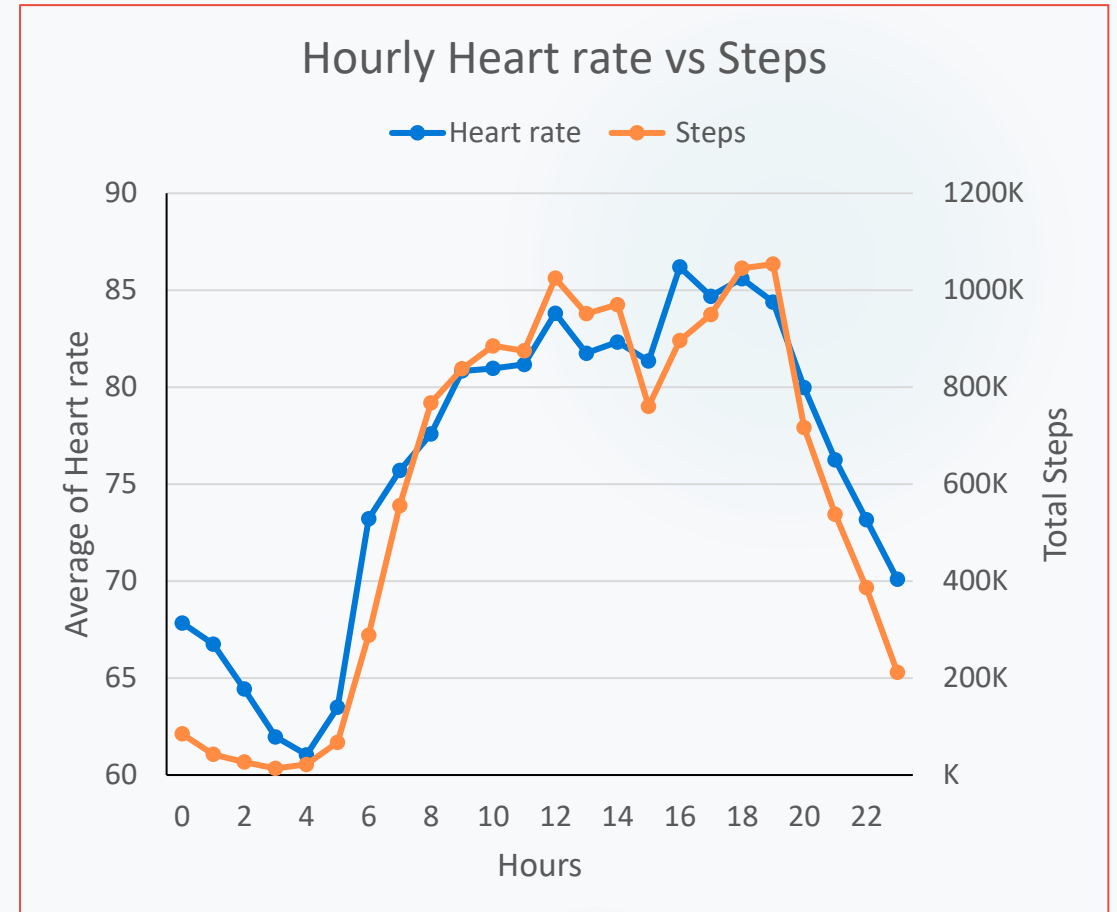
Hourly Trend of Average Heart Rate and Calories Burned

- ▶ Calories burned follow the same trend as heart rate throughout the day.
- ▶ Peak calorie burn happens between 5 AM – 9 AM and 5 PM – 6 PM, matching high activity periods.
- ▶ Lowest calorie burn occurs between 12 AM – 4 AM, indicating resting hours.



Hourly Trend of Average Heart Rate and Total Steps

- ▶ Heart rate rises in sync with increased step counts.
- ▶ Step activity peaks twice → 12AM & 7AM, coinciding with peak activity times.
- ▶ Lowest activity is seen between 12 AM and 4 AM, matching resting periods.



Recommendations

1. Sleep Improvement 🛏️

- ▶ Educate users on the ideal 6–8 hours of sleep.
- ▶ Introduce bedtime reminders and sleep tracking insights.

2. Weight & BMI Management ⚖️

- ▶ Suggest personalized fitness goals based on BMI categories.
- ▶ Offer guided workout plans for overweight users.

3. Increase Engagement 📅

- ▶ Add gamification features (step challenges, badges).
- ▶ Send weekly summaries to improve long-term usage.

4. Health Optimization ❤️

- ▶ Highlight abnormal heart rate patterns for early detection.
- ▶ Encourage micro-breaks during long sedentary periods.

Conclusion

- ▶ Users are moderately engaged, but tracking consistency can improve.
- ▶ Sleep patterns are mostly healthy, but 30% of users are sleep-deprived.
- ▶ Steps, calories, and heart rate peak during midday and evenings.
- ▶ Targeted wellness campaigns and personalized plans can improve overall health outcomes and device engagement.